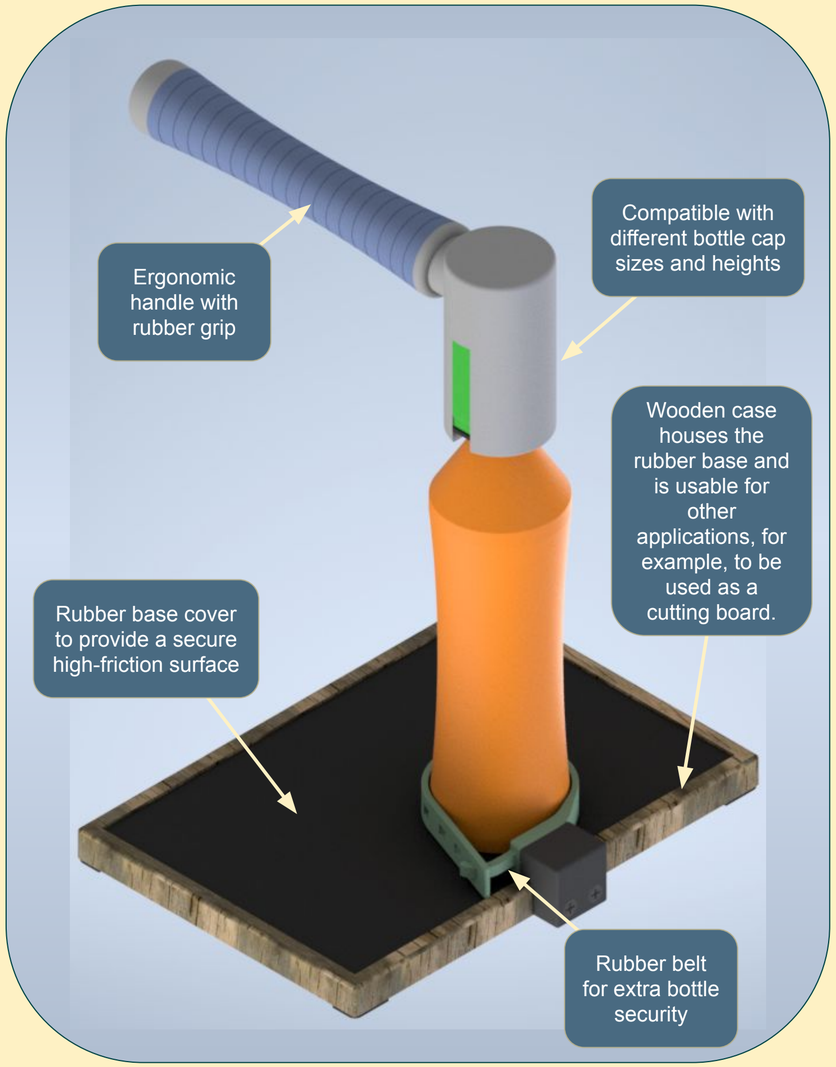
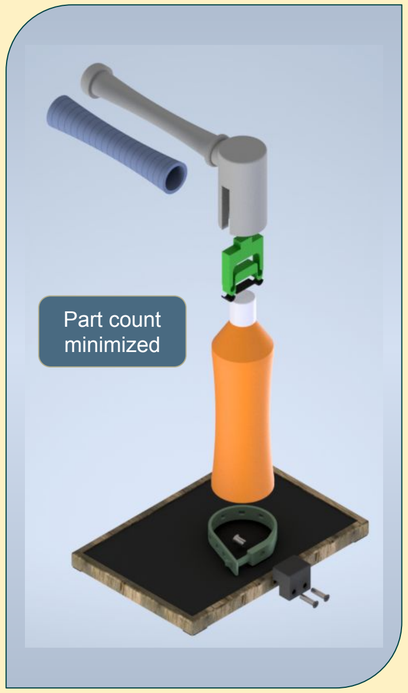




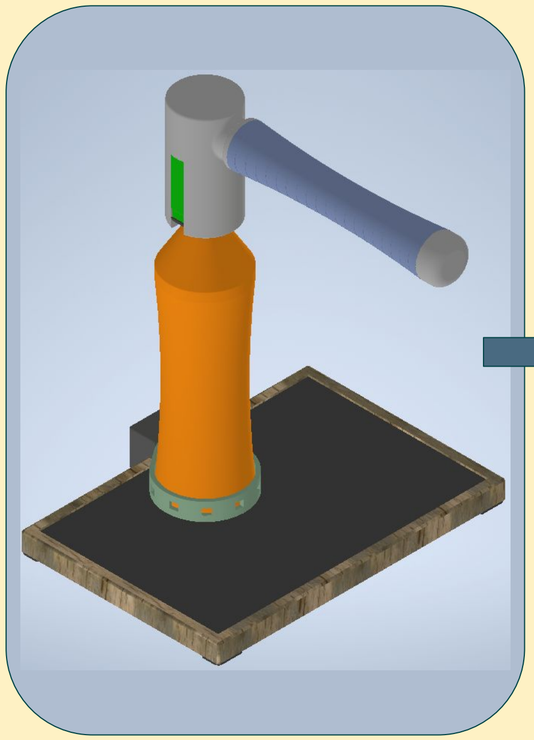
How it works

These illustrations show the final design, it incorporates the use of a **compliant mechanism** that takes advantage of the **malleability** of plastic. This device is very **simple to use**, one must only **place** the top part of the mechanism on to the bottle and the compliant mechanism **grips** the bottle once a **slight force** is applied by the user. The handle is then **turned**, which provides a **mechanical advantage** by providing a **larger torque** due to its length. The handle has an **ergonomic sleeve** to provide **comfort** to the user and a **multipurpose** base is provided to secure the bottle and hold it still.

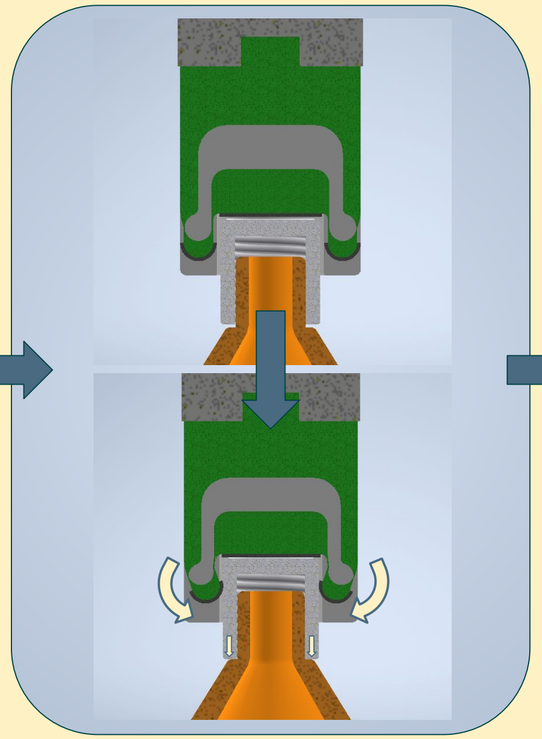
The final design was based on the first concept, the core ideas and mechanisms were revised and improved. The compliant mechanism arms were changed to half circles to increase stability, contact surface and grip reliability. A rubber layer was added to the arms and on the bottom of the mechanism to increase grip onto the bottle. The handle's ergonomics were improved to make gripping the device more comfortable and a soft rubber sleeve was added to make the handle softer.

The base was completely re-designed, it now has a rubber surface that produces a lot of friction with the bottle, eliminating the need for a gripping system. For redundancy, a rubber belt is bolted onto the side of the base and can be used to secure the bottle further.

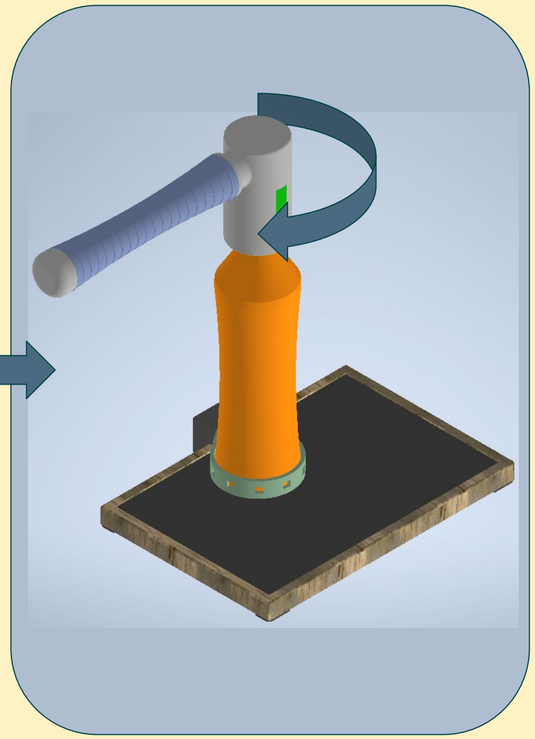
Storyboard



Bottle is placed on the base and the bottle cap opener is placed on the bottle cap. Tightening the belt around the bottle base is optional.

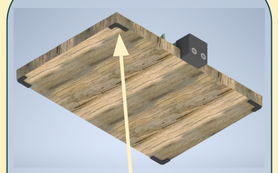


Bottle cap cover is pushed down and the compliant mechanism grips onto the bottle cap. This also releases the child-proof mechanism.



The bottle cap cover (along with the cap) are twisted with the handle and the bottle is opened.

Requirements	✓ / ✗
1) Performance	
Provides mechanical advantage when opening the bottle	✓
Can be operated individually	✓
Simple and easy to use	✓
Works for different bottle sizes	✓
Affordable	✓
2) Geometry	
Light weight	✓
Minimised number of components	✓
Easy to store	✓
3) Safety	
No sharp edges	✓
Safe to use	✓
Comfortable to use	✓
4) Material	
Comfortable handle	✓
Eco-friendly	✓
5) Ergonomics	
Ergonomically sound	✓



Potential Design Improvements:

The top could be changed to make it more multi-purpose. The mechanism could be improved to only require one hand to open the bottle.